



Advanced Laboratory Introductory Session 2025/26

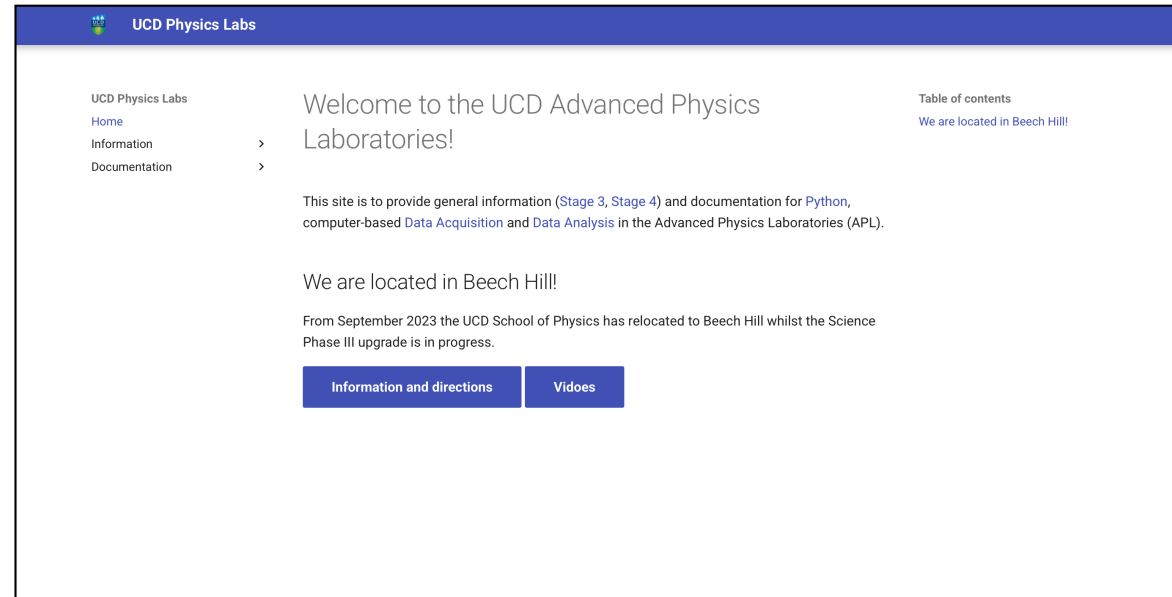
APL Stage 4 Introductory Session

9th September 2025

Welcome back!



Advanced Laboratory Introductory Session 2025/26



Advanced Laboratories are located in Beech Hill on the ground and first floors of the B-wing:

- Rooms B001, B002, B003, B004 and B005 (Ground Floor)
- Rooms B103 (Paul Hanratty & Joe Branson's Offices) and B105 (First Floor)

Reminder: When accessing the labs on the first floor, do so using the stairs at the back of the labs on the ground floor to avoid going through classroom in B106-BH while lectures are taking place!



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No Physics Library or common room in Beech Hill.

You will be able to use B000–BH as your base:

- Microwave and kettle available at end of room
- Must make sure that room is kept clean / no strong smells
- At end of day, chairs must be left ready for next day's lectures

Please let us know if there are any issues with schedules and commuting between UCD main campus and Beech Hill.





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Points of contact:

- Main Office (B109–BH)
- Bairbre Fox, School Manager (A109–BH)
- Prof. Dominic Zerulla, HOS Office (A108–BH)
- Assoc. Prof. León Vitró, Stage 4 teaching team coordinator (A002–BH)
- Prof. Quinn, Dr. McCormack and Assoc. Prof. Mitchell (Programme Directors for PASS, SH and TP)





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The following laboratories are Year Long modules (run over Autumn and Spring trimesters):

PHYC 40970: Advanced Lab. II (15 ECTS)

PHYC 40600: Astronomy & Astrophysics Lab II (15 ECTS)

The following laboratories must be completed in a single trimester:

PHYC 40330: Advanced Laboratory for TP III (5 ECTS)

PHYC 40690: Advanced Physics Laboratory (5 ECTS)



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Advanced Physics Laboratory Philosophy:

- Educational environment.
- Aimed to develop your critical thinking and problem-solving skills.
- Not only about report, but also demonstrating understanding of the experiment, including set-up, execution and data analysis.
- Interaction with staff & demonstrators is important.
- Work must be your own (plagiarism) – if a report is found to be plagiarised from another one, this will be investigated and could result in **both** reports being marked as zero.

Everything you include should be your own work. ChatGPT should not be used for the laboratories. You must be able to explain and defend everything written in your reports (and code!) during the interviews with staff. Your ability to do so will be part of the grading for the experiment.



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WEBSITE URL FOR ADVANCED PHYSICS LABORATORY

Make use of the APL website – let us know of any issues!

<https://physicslabs.ucd.ie>

- General information
- Deadlines
- Slides of presentations
- Python programming
- Data acquisition
- Data analysis

UCD Physics Labs

Stage 4 (2025-2026)

UCD Physics Labs
Home
Information
Beech Hill
Stage 3
Stage 4
Dignity and Respect
Documentation

Introductory Lecture

The Stage 4 Advanced Laboratory Introductory Talk will be given by Assoc. Prof. Luis Leon-Vintro at 2pm on Tuesday 9 September 2025 in room B000-BH (Beech Hill).
Slides from the Introductory Lecture by Assoc. Prof. Vintro are available [here](#).

Report-Writing Refresher Session

Dr. Tom McCormack will conduct a report-writing refresher session on Wednesday 17 September at 4pm in room B106-BH (Beech Hill).

Advanced Laboratory Hours and Dates

Attention!

All students are required to sign in and out of the lab using the Google form accessed via the QR code posted in the labs.

The laboratories are open with demonstrator and staff support from 2 pm to 6 pm on Tuesday, Wednesday and Thursday afternoons.

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PHYC 40330 and PHYC 40690 (5 credit, one-trimester)
Report Submission Deadlines
PHYC 40600 and PHYC 40970 (15 credit, two-trimester)
PHYC 40330 and PHYC 40690 (5 credit, one-trimester)
Late Submission of Laboratory Reports

Note: New software for 2025–26 – please let us know if you find a broken link, etc.



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Workload:

- PHYC 40970/40600: 340 hrs (~12 hrs/week)
- PHYC 40330/40690: 110 hrs (~6 hrs/week)

Laboratory hours:

- Tuesdays, Wednesdays & Thursdays 2–6 pm
Expected to be present at assigned times and welcome to attend outside time-tabled hours.

Staff:

Module coordinators:

- Prof. Quinn (PHYC40600)
- Dr. McCormack (PHYC40970)
- Assoc. Prof. León Vintró (PHYC40330)

Other staff involved:

- Dr. Hayden
- Assoc. Prof. Lobaskin, Dr. Burgess (Computational)
- Assoc. Prof. Vohnsen, Dr. Benedetto

Technical staff:

- Mr. Hanratty, Dr. Branson



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John
Quinn



Luis
León Vintró



Tom
McCormack



Paddy
Hayden



Antonio
Benedetto



Vladimir
Lobaskin



Brian
Vohnsen



Andrew
Burgess



Joe
Branson

Paul
Hanratty



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Schedule (strict deadlines for each trimester) :

PHYC 40970: (SH)	Trimester 1: 4 experiments Trimester 2: 2 experiments
PHYC 40600: (PASS)	Trimester 1: 4 experiments Trimester 2: 2 experiments
PHYC 40330/40690 (TP)	Trimester 1/2: 2 experiments

For PHYC40970 and PHYC40600, Reports 1 & 2 must be submitted by 5pm on Tuesday, 28th October. Report 3 and all experimental work for experiment 4 must be completed before end of term. Report 4 due by 5pm on Monday 19th January 2026. Reports 5 & 6 due by 5pm on Monday 9th March 2026 (Labs close on Thursday 26th February).

This does not apply to PHYC40330/40690, which are trimester modules. For these, reports 1 & 2 due by 5pm on Monday 10th November (Trimester 1) or 5pm on Monday 30th March (Trimester 2). For trimester 1, Labs close on Thursday 30th October.



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Remember that there you will need ~3 weeks per experiment.
Manage your time wisely and do not leave things to the last week!



Trimester 1

Tuesday 9th Sept – **Thursday 27th November** (Labs close PASS/SH)
Stage 4 student presentations start Wednesday 1st October
(Week 4) Dr. McCormack will contact with details on format,
schedule, etc... (scheduled 2–3pm on Wednesdays)

***Thursday 30th October** – Lab closes for PHYC40330/40690*
All reports due Monday 10th November

Trimester 2

Labs re-open: Tuesday 20th January

Labs close for PASS/SH students: **Thursday 26th February**

All reports (PASS/SH) due Monday 9th March

***Friday 13th March** – Lab closes for PHYC40330/40690*
All reports due Monday 30th March



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Reports:

- All reports are to be typed (do not forget to append Jupyter notebooks and program code to the end of your report – Be careful with Jupyter Notebook PDF appendices and make sure that are not all on one page!)
- An electronic version of each report in a single pdf document is to be submitted through Brightspace (originality checker will run on it) and a copy emailed to Paul Hanratty.
- Please remember to include an ‘Analysis’ section in your report and do not relegate it to an undocumented Python appendix!
- Each experiment and report to be completed (submitted) before starting next one.
- We do not need large reports – brief, factual, informative reports which are clear and understandable are preferred.



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Late Submission of Laboratory Reports:

The UCD Late Submission of Coursework policy will apply:

- Coursework submitted up to 10 working days past the due date will be graded but a grade penalty will be applied.
- Coursework submitted at any time up to and including 5 working days after the due date will have the grade awarded reduced by one grade (e.g., from B– to C+).
- Coursework submitted more than 5 working days but up to and including 10 working days after the due date will have the grade awarded reduced by two grade points (e.g., from B– to C).
- Coursework received more than 10 working days after the due date will not be accepted or graded.



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Requirements:

- Statement of intent must be completed before experiment and submitted electronically to Paul and staff member in charge of the experiment
- Keep records of experiments in a dedicated notebook
- Keep a copy of all files relating to laboratory work (must be available at end of year) – make frequent backups to avoid data loss!
- Sign in and out (Google form via the QR codes posted in labs)
- All reports should be typed and submitted **electronically** as a single pdf file (do not forget to append all iPython notebooks and program code to the end of reports) through *Bright Space* and a copy emailed to Paul Hanratty
- Complete experiment (and report) before starting next one
- If there is no activity on an experiment for two weeks, the experiment might be allocated to someone else



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Statement of Intent (SOI)



Statement of Intent

UCD School of Physics
Advanced Laboratories

Name:

Experiment:

Date:

The Statement of Intent should address the following four points about the experiment you are about to undertake:

1. What is the aim of the experiment?
2. What measurements will be made and how?
3. How will the final result be obtained from the experimental data?
4. What are the main safety concerns with the experiment and precautions that should be taken?

The statement should be reviewed by a member of staff before you begin the experiment.

Motivation:

- While writing reports students realise what they should have done
- Know what you have to do in advance and make a plan!
- Safety

Equipment:

- You should not use any equipment until you have discussed experiment with a member of staff
- It is ok (expected) to look at the equipment before writing SOI. Handout not always necessarily reflect the equipment set up!

The IOP requested student SOI as part of their accreditation process



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Laboratory Safety

- Safety in the lab is of the utmost importance
- People
- Equipment
- Safety concerns need to be thought about in advance and included in the SOI.
- Please think critically about the safety aspects before using any equipment (ongoing).
- Equipment: we frequently have blow fuses – please know how to wire up a basic circuit including how to put an ammeter in a circuit!
- Any experiment requiring the use of LN2 will need prior training in how to handle cryogenic liquids.



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Assessment:

- Commitment, quality of work, understanding & practical skills
- Experiments are designed to be extendable – show initiative in pursuing experiments beyond specified instructions
- Assessment based on reports and staff judgement + discussions after each experiment (to be arranged with staff)

Assessment breakdown: 80% laboratory report and discussion with staff; 20% for general continuous assessment mark by the staff

Please remember that one experiment is worth 2.5 ECTS (50–75 hr of effort). This is 50% that of a typical lecture module. Budget your time and effort accordingly!



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Other information:

- There will be a refresher 'Report Writing' session on Tuesday 16th September (details TBC)
- Class reps for each degree programme (SH, PASS, TP) for Teaching Team Meetings (one meeting per semester)
- Behaviour – treat labs as work environment
- Keys for lockers / date to be arranged in the coming days
- You should have received your first assigned lab. If not, please contact Paul Hanratty.



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OTHER IMPORTANT INFORMATION:

- Labs are not open on Mondays and Fridays.
- SH Projects done in separate module (PHYC40960).
- Laboratory report discussion held after each experiment (same as in Stage 3).
- Website url for the labs: <https://physicslabs.ucd.ie>
- Google Form sign in and sign out via the QR codes posted in the lab – required for attendance
- New email notification list (one way from staff to students) UCD-PHYSICSLABS4@listserv.heanet.ie – contact person: J. Quinn



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OTHER IMPORTANT INFORMATION:

- While in the lab, try make the best use of your time – get the experiment set up and get the data in a planned and efficient way.
- Work from first week of term and do not leave experiments to the end!
- Please keep in contact and enjoy the labs.



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- The Advanced Physics Laboratories aim to provide an educational environment with the highest standards of behaviour and ethics, free of bias and any forms of harassment or bullying.
- You should familiarise yourself with and abide by the UCD policies on Student Conduct and Academic Integrity (<https://www.ucd.ie/secca/studentconduct/>)

UCD Dignity and Respect Support Service

Supporting any student, employee or UCD community member who is impacted by issues of **bullying, harassment or sexual misconduct**

- Confidential and proactive support
- Specialist information and guidance
- Support to explore informal and formal resolution options
- Ongoing aftercare support
- Drop in or make an appointment

L532, Level 5 James Joyce Library Building

respect@ucd.ie ucd.ie/dignityandrespect



**LABS ARE NOW OPEN.
START WORKING TODAY!**